

Photovoltaic layout, shading and storage | Models and formulas

Technical note revised for the PV Layout application, 28 September 2026. Based on the mathematical appendix by Alexandre Littardi, 25 September 2026; revised against the current source code.

What was corrected

- Dated simulations now use their actual calendar year and 365 or 366 days, rather than a fixed 2024/365 assumption. The undated Shadow preview still uses representative year 2024.
- The earlier note only described clear-sky shading. The annual Energy/BESS simulation can use imported historical weather; a chosen weather source is recorded with the results.
- The solar-to-module relation is a simplified irradiance and linear temperature model. It does not solve a single-diode circuit or reproduce a complete NOCT/SAM thermal model.
- DC voltage drop uses the lengths of both conductor runs, A+B. Replacing A+B by 2L is valid only when the two runs have equal length.
- The original MPPT voltage interval described a spread of string voltages, not electrical compatibility. Parallel strings of unequal lengths are flagged for review.
- The updated note includes hourly direct PV use, shared BESS charge/discharge power limits, SOC constraints, export limit, curtailed energy and net grid cost.
- DC conductor sections, protective devices, cold-temperature voltage, cable thermal rating, battery expansion and grid approval still require certified equipment information and engineering review.

Reading the equations

The expressions below describe the code paths used by this version. A missing weather profile selects the clear-sky estimate. Units, direction conventions and limits must be read with each equation. The guides in Home show the same material in an interactive form.

Symbols and scope

Angles are in degrees unless specified. Latitude phi is north-positive, longitude lambda east-positive, local hour h uses the supplied UTC offset u. G is W/m², P is W, E is Wh. Calculations are estimates, not equipment certification.

$$\gamma = \frac{2\pi}{N_y} \left(n - 1 + \frac{h-12}{24} \right)$$

Fractional year: N_y is 365 or 366 for the calculation year; n is the day number. The undated preview uses 2024.

Solar position

Solar elevation is corrected for refraction, while azimuth uses geometric elevation. Shadow preview uses representative year 2024 because its controls give only day/month; dated simulations and annual Energy/BESS use the real year. The annual routine derives the UTC offset for each hourly Europe/Rome timestamp.

$$E_t = 229.18(0.000075 + 0.001868\cos \gamma - 0.032077\sin \gamma - 0.014615\cos 2\gamma - 0.040849\sin 2\gamma)$$

Equation of time in minutes (Spencer/NOAA approximation).

$$\delta = 0.006918 - 0.399912 \cos \gamma + 0.070257 \sin \gamma - 0.006758 \cos 2\gamma + 0.000907 \sin 2\gamma - 0.002697 \cos 3\gamma + 0.001480 \sin 3\gamma$$

Solar declination in radians.

$$\omega = \frac{60h + E_t + 4\lambda - 60u}{4} - 180$$

Hour angle: negative in the morning, positive in the afternoon.

$$\alpha_{geo} = \arcsin(\sin \phi \sin \delta + \cos \phi \cos \delta \cos \omega)$$

Geometric solar elevation; trigonometric arguments are converted to radians.

$$\alpha = \alpha_{geo} + \frac{1.02}{60 \tan(\alpha_{geo} + 10.3/(\alpha_{geo} + 5.11))}$$

Sæmundsson refraction for geometric elevation greater than -0.85° . The tangent argument is converted to radians.

$$\cos A = \frac{\sin \delta - \sin \alpha_{geo} \sin \phi}{\cos \alpha_{geo} \cos \phi}$$

Solar azimuth from north: arccos in the morning, 360° minus arccos in the afternoon.

Irradiance and module power

A simplified clear-sky scenario is used for the shadow study. The Energy/BESS annual simulation can instead use imported historical plane-of-array irradiance AND hourly ambient temperature for a single module orientation. The 20°C ambient assumption and constant 0.20 albedo apply to the clear-sky model only.

$$m_a = \frac{1}{\sin \alpha}, \quad G_{DNI} = 1367(0.7)^{m_a^{0.675}}$$

Clear-sky direct normal irradiance; set to zero at solar elevation at or below 0.5° .

$$c_i = \max(0, \min(1, \sin \alpha \cos \beta + \cos \alpha \sin \beta \cos(A - \gamma_p)))$$

Incidence factor for module tilt beta and module azimuth gamma_p.

$$G_{POA} = G_{DNI} c_i + 0.12 G_{DNI} \sin \alpha \frac{1 + \cos \beta}{2} + 1.12 G_{DNI} \sin \alpha 0.20 \frac{1 - \cos \beta}{2}$$

Direct + simplified sky diffuse + ground reflection, with nonnegative sine of solar elevation.

$$G_{eff} = G_{POA}(1 - f_s), \quad T_c = T_{ambient} + (T_{NOCT} - 20) \frac{G_{eff}}{800}$$

Effective irradiance after fractional shading. $T_{ambient}$ is 20°C for the clear-sky calculation; it comes from the hourly weather profile when supplied.

$$P = P_{STC} \frac{G_{eff}}{1000} \max(0, 1 + \frac{\mu_p}{100}(T_c - 25))$$

Linear power estimate (muP in $\%/^\circ\text{C}$), not an I-V or bypass-diode calculation.

Time, shading and layout

A shadow footprint is a square at the configured obstacle position. Its projection and polygon intersection are computed separately for each roof height. Real roof clearances and structural constraints are outside the grid model.

$$E = \sum_{k=1}^M P(t_{k,mid}) \Delta t_k, \quad E_{loss} = \max(0, E_{unshaded} - E_{shaded})$$

Midpoint integration; E is Wh for P in watts and time in hours.

$$E_{string} = \sum_j E_j, \quad r_{loss} = 100 \frac{\sum_j E_{loss,j}}{\sum_j E_{unshaded,j}}$$

Aggregated module energies and relative estimated string losses. This is not an electrical current-limited string model.

$$Spread = \max_j(r_{loss,j}) - \min_j(r_{loss,j})$$

Difference between the most and least affected modules of one string; a mismatch indicator only.

$$A = \frac{1}{2} \left| \sum_{i=1}^p (x_i y_{i+1} - x_{i+1} y_i) \right|$$

Shoelace area for a simple polygon; its intersection with a panel is obtained through polygon clipping.

$$d = \frac{H_{obstacle} - H_{roof}}{\tan \alpha}, \quad f_s = \min\left(1, \frac{A_{intersection}}{A_{panel}} o\right)$$

Shadow length at positive elevation and positive height difference; o is obstacle opacity in [0,1].

$$n_{col} = \lfloor \frac{W}{w} \rfloor, \quad n_{row} = \lfloor \frac{H}{h} \rfloor$$

Panel grid before user edits; alignment uses residual width/height, and rotation acts about each panel centre.

$$o_x \in \{0, (W - n_{col}w)/2, W - n_{col}w\}, \quad o_y \in \{0, (H - n_{row}h)/2, H - n_{row}h\}$$

Left/centre/right and top/centre/bottom alignment of the residual clearance.

$$x' = c_x + (x - c_x)\cos\theta - (y - c_y)\sin\theta$$

Rotation of x around each panel centre.

$$y' = c_y + (x - c_x)\sin\theta + (y - c_y)\cos\theta$$

Rotation of y. The convex hull and polygon clipping use Andrew and Sutherland-Hodgman algorithms.

Strings, routing and voltage drop

N modules in series multiply voltage but retain module current. Parallel MPPT strings add currents; unlike a loose voltage interval, differing string voltages need an explicit mismatch check. Routes A and B are separate physical conductor paths.

$$P_{string} = \frac{NP_{STC}}{1000}, \quad V_{oc,string} = NV_{oc}, \quad V_{mpp,string} = NV_{mpp}$$

kWp and voltages at STC; Isc and Imp each equal the module current for a series string.

$$I_{MPPT} = \sum_{j=1}^k I_{string,j}, \quad P_{inv,DC} = \sum_{s \in inv} P_{string,s}$$

Parallel current and assigned DC peak power. Check maximum voltage at cold temperature separately.

$$L_{loop} = L_A + L_B, \quad \Delta V = \frac{I \rho L_{loop}}{S}$$

Two conductor lengths including modeled vertical drop and reserves; rho is $\Omega \cdot \text{mm}^2/\text{m}$, S is mm^2 .

$$L_{orthogonal} = |x_2 - x_1| + |y_2 - y_1|$$

Fallback Manhattan route between two points without a drawn path; the drawn cable network uses Dijkstra shortest-path search.

$$V_{oc, cold} = NV_{oc, STC} \left[1 + \frac{\mu_{Voc}}{100} (T_{min} - 25) \right]$$

Cold-voltage estimate where the module voltage temperature coefficient and site design temperature are supplied; otherwise the check remains open.

$$S_{min} = \frac{I \rho (L_A + L_B)}{U \epsilon_{target} / 100}, \quad \epsilon_{actual} = 100 \frac{I \rho (L_A + L_B)}{S U}$$

Equivalent to 2L only if $L_A = L_B = L$. Sizing is voltage-drop-only; no ampacity or fault study.

$$I_{AC, nom} = \frac{1000 S_{AC, kVA}}{\sqrt{3} U_{LL}}$$

Estimated three-phase line current from apparent inverter rating and entered line voltage; this is not a breaker selection.

Energy / BESS: inputs

Source: self_consumption.py, production_from_program(). Index h denotes a one-hour local-time slot (Delta t = 1 h); energies are kWh and powers kW. Consumption is imported as 24 columns per local day. Missing load values are estimated from up to seven preceding days at the same clock hour and flagged. PV is evaluated at the hour midpoint, using clear-sky irradiance or imported historical plane-of-array irradiance and ambient temperature. Historical weather currently supports one roof orientation. The 24-slot convention approximates daylight-saving transition days.

$$E_{PV, h} = \min \left(P_{AC, cap}, f_{AC} \frac{\sum_i P_{i, h}}{1000} \right) \Delta t$$

$P_{i, h}$ is the shaded module power in W, as defined in the module model. f_{AC} is the additional conversion/loss factor (default 0.90). $P_{AC, cap}$ is the aggregate configured inverter cap. The code uses the numerical kVA ratings for this kW cap, assuming unity power factor; confirm active-power ratings before relying on clipping estimates.

$$\hat{E}_{load, h} = \frac{1}{|D_h|} \sum_{d \in D_h} E_{load, h-24d}$$

D_h contains available preceding days $d = 1, \dots, 7$. An error is raised if no prior value exists. Estimated entries must be reviewed separately from measured consumption.

Energy / BESS: dispatch

Source: battery_dispatch.py, simulate_bess() and simulate_three_options(). The same PV and load inputs are used for PV only, one cabinet and two cabinets. Default cabinet capacity is 522.496 kWh. Two cabinets double capacity, while plant charge/discharge caps remain 250/150 kW. These are study settings. There is no grid charging, battery export or tariff-based scheduling. SOC starts at its configured minimum; the end SOC is not forced back to the start.

$$E_{direct,h} = \min(E_{PV,h}, E_{load,h})$$

PV first supplies the load directly.

$$E_{surplus,h} = E_{PV,h} - E_{direct,h}, \quad E_{deficit,h} = E_{load,h} - E_{direct,h}$$

Only surplus PV can charge storage; only the residual load can receive a discharge.

$$C_n = nC_1, \quad Q_{min} = C_n s_{min}, \quad Q_{max} = C_n s_{max}$$

For $n = 1$ or 2 cabinets, Q is stored energy (kWh) and s is a fractional SOC. Default $s_{min} = 0.10$ and $s_{max} = 0.90$. PV-only has zero charge and discharge.

$$\eta_c = \eta_d = \sqrt{\eta_{rt}}, \quad \eta_{rt} = 0.90$$

Charge and discharge efficiencies are split equally; the round-trip efficiency is configurable.

$$E_{charge,h} = \min(E_{surplus,h}, P_c \Delta t, \frac{Q_{max} - Q_h}{\eta_c})$$

AC energy sent to the battery, constrained by surplus, charge power and available capacity.

$$Q_h^+ = Q_h + \eta_c E_{charge,h}$$

Stored energy after the charge step.

$$E_{discharge,h} = \min(E_{deficit,h}, P_d \Delta t, (Q_h^+ - Q_{min}) \eta_d)$$

AC energy delivered by the battery to the load. Surplus and deficit cannot both be positive in the same hourly slot.

$$Q_{h+1} = Q_h^+ - \frac{E_{discharge,h}}{\eta_d}, \quad SOC_{h+1} = 100 \frac{Q_{h+1}}{C_n}$$

SOC shown on the hourly chart is the end-of-hour state. The code clips numerical round-off to the SOC limits.

Energy / BESS: grid

Source: `self_consumption.py`, `balance()`, and `battery_dispatch.py`. The requested export cap is 300 kW and is not recorded as approved. The existing 507 kW contract is stored separately; it does not replace the export cap or constrain grid imports in this dispatch model.

$$E_{grid,h} = E_{deficit,h} - E_{discharge,h}$$

Grid purchases cover the remaining load, including hours when PV output is zero.

$$E_{export,h} = \min(E_{surplus,h} - E_{charge,h}, P_{export} \Delta t)$$

Remaining PV surplus is exported up to the configured export cap.

$$E_{curtailed,h} = E_{surplus,h} - E_{charge,h} - E_{export,h}$$

Surplus that cannot be consumed, stored or exported is counted as curtailed PV.

$$E_{load,h} = E_{direct,h} + E_{discharge,h} + E_{grid,h}$$

Load-side energy conservation, checked for annual totals.

$$E_{PV,h} = E_{direct,h} + E_{charge,h} + E_{export,h} + E_{curtailed,h}$$

PV-side energy conservation. PV before grid curtailment is the reported production quantity.

Energy / BESS: metrics

Source: battery_dispatch.py annual aggregation; mixins/hourly_chart_ui.py displays the results. Monthly and annual values are sums of the hourly rows. The annual chart shows monthly totals in MWh. The hourly chart shows PV/load curves, stacked load sources and battery SOC. Grey bands mark zero PV production. Percentages use the totals below, not the average of hourly percentages.

$$E_{useful} = \sum_h (E_{direct,h} + E_{discharge,h})$$

Useful PV supplied to the load, directly or through storage.

$$E_{onsite} = \sum_h (E_{direct,h} + E_{charge,h})$$

PV retained on site includes energy entering storage; it is not the same quantity as useful load served.

$$L_{battery} = \sum_h (E_{charge,h} - E_{discharge,h}) - (Q_{end} - Q_{start})$$

Battery losses deduct the change in stored energy, avoiding a false loss attribution to the final SOC.

$$SelfSufficiency = 100 \frac{E_{useful}}{\sum_h E_{load,h}}$$

Percentage of refrigerator consumption served by PV and BESS. The displayed value is zero if the load denominator is zero.

$$PVRetained = 100 \frac{E_{onsite}}{\sum_h E_{PV,h}}, \quad UsefulPV = 100 \frac{E_{useful}}{\sum_h E_{PV,h}}$$

PV kept on site and useful self-consumption as shares of production. Zero production gives zero displayed ratios. Neither rate should be interpreted as storage efficiency.

Energy / BESS: costs

Source: energy_economics.py, evaluate_options(). All energies below are annual totals. p_b and p_s are purchase and export prices in EUR/kWh. The defaults are p_b = 0.25 (0.32 can also be selected), and two export-price cases p_s = 0.07 and 0.08. PV investment is EUR 250,000; each cabinet adds EUR 50,000. Fixed charges, taxes, maintenance, financing, degradation and replacement are not included.

$$C_0 = p_b E_{load}, \quad C_{net,n} = p_b E_{grid,n} - p_s E_{export,n}$$

Baseline purchase cost without PV, and annual net electricity cost for scenario n.

$$B_n = C_0 - C_{net,n} = p_b E_{useful,n} + p_s E_{export,n}$$

Annual benefit includes both avoided purchases and export revenue.

$$K_n = K_{PV} + n K_{BESS}, \quad T_n = \frac{K_n}{B_n}$$

Simple whole-project payback in years, only when B_n > 0; otherwise no positive payback is displayed.

$$\Delta B_n = B_n - B_{n-1}, \quad T_{extra, n} = \frac{K_{BESS}}{\Delta B_n}$$

Incremental payback compares one cabinet with PV-only, then two cabinets with one. It is displayed only if the incremental benefit is positive; foregone export revenue is already deducted.

Energy / BESS: EMS

Source: `energy_economics.py`, `storage_margin_per_charge_kwh()`. This window evaluates the fixed self-consumption dispatch. It sends no control commands and does not optimize time-varying prices, grid arbitrage or future load forecasts. Recalculate energy after changing source profiles, layout or storage. Updating prices alone recalculates costs from the existing hourly energy results.

$$m_{charge} = \eta_{rt} p_b - p_s$$

Gross incremental value of one PV kWh charged and later delivered to the load instead of exported now. Assumes available future demand and SOC headroom; excludes ageing and auxiliary consumption.

$$p_{s, break} = \eta_{rt} p_b$$

Export price at which that simplified gross storage margin is zero. This indicator does not alter the current dispatch algorithm.

Traceability and limits

- Solar geometry and irradiance: `mixins/shadow_energy.py`; obstacle footprint, clipping and area: `mixins/shadow_geometry.py`; panel grid: `mixins/zone_tools.py`.
- String and MPPT STC values: `mixins/diagram_tools.py` and `detailed_electrical.py`; network routes and two DC terminals: `mixins/cable_network.py` and `mixins/two_pole_cables.py`; voltage-drop recommendation: `mixins/notes_tools.py`.
- Hourly generation: `self_consumption.py`; one/two BESS dispatch: `battery_dispatch.py`; avoided purchases, export revenue and payback: `energy_economics.py`.
- The roof shadow model uses a square obstacle footprint and separate zone heights, not an as-built mesh of a lattice tower. String mismatch, diodes, snow, wind, bifacial gain and all network protection settings are outside this simulation.
- Public reference: NOAA General Solar Position Calculations, <https://gml.noaa.gov/grad/solcalc/solareqns.PDF> (366-day leap-year denominator).
- Public reference: Sandia PV Performance Modeling Collaborative, NOCT Cell Temperature, <https://pvpmc.sandia.gov/modeling-guide/2-dc-module-iv/cell-temperature/noct-cell-temperature/> (the application uses a simplified ambient-temperature estimate).

Interpretation for the 516-module project

- The supplied 516-module reference represents 371.52 kWp of nominal DC modules assigned to three 125 kVA hybrid inverters and thirty-one strings. The stored annual clear-sky calculation is a scenario, not measured PV yield; historic weather and load data must be checked for matching dates and time conventions.
- One MC-L522 has two battery clusters connected to two inverter BAT ports in the available DEYE drawing. The third hybrid is dedicated to PV. An additional MC-L522 cabinet must not be represented as approved parallel DC wiring until a manufacturer-approved extension drawing is available.
- The requested 300 kW export setting is a scenario limit for the hourly dispatch. The 507 kW existing contract is a different quantity. Neither value by itself documents a granted injection authorization or a tested grid-interface protection.
- Cable paths are estimates until traced on site. A valid A+B record includes two conductor routes, roof height difference and configured reserve. Conductor sizing from DC voltage drop does not substitute for ampacity, short-circuit, thermal and protective coordination studies.
- Shadow preview and dated shadow studies accept a manually supplied UTC offset; the undated preview uses representative year 2024. Dated shadow simulations use the real calendar year. The annual simulation derives hourly offsets from Europe/Rome and its timestamp. Clear-sky module temperature assumes 20°C ambient; imported weather supplies hourly ambient temperature and plane-of-array irradiance for a single module orientation.